



Scout

A purpose-aware company intelligence agent

Give it a company and a reason. It plans research around that reason, cites every claim, checks its own work, and gets faster with every run.

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AI Agentic System Challenge · September 2026

github.com/UditSinghChauhan/scout-agent

Replay of recorded run 20260927-155410-bc87, recorded 2026-09-27 15:54:10+05:30

Goal: We sell a campus hiring-challenge platform. Research Zoho as a prospect and tell us how to pitch.

Cold start: no stored facts about Zoho yet.

intake: done recall: done plan: done execute: done synthesize: done verify: done reflect: done

Plan · Sales prospect

done	1. What is Zoho's overall company size, headquarters location, and global office footprint?
unknown	2. How many university recruitment programs, campus hiring events, or graduate-entry roles does Zoho run annually, and which campuses are targeted?
done	3. What talent acquisition tools, assessment platforms, or hiring-challenge solutions does Zoho currently use or mention in its tech stack?
unknown	4. Who are the senior decision-makers responsible for campus recruiting, talent acquisition, or early-career hiring at Zoho (roles only, not names)?
unknown	5. What recent news, product launches, funding rounds, or strategic initiatives has Zoho announced that could create a buying signal for a campus hiring-challenge platform?

Trace

> Step 1: done · 2 tool calls · critic: complete
> Step 2: unknown · 1 tool call · critic: unknown
> Step 3: done · 3 tool calls · critic: complete
> Step 4: unknown · 3 tool calls · critic: unknown
> Step 5: unknown · critic: unknown
> Run events (?)

Score	Tool calls	LLM calls	Tokens	Seconds	Citation coverage	Memory steps	Provider switches
Fit 70/100	9	35	38,866	172.8 s	100%	0	3

Brief

Zoho – Campus Hiring-Challenge Platform Pitch

Purpose: Sales prospect · Run · 20260927-155410-bc87

Score: Fit 70/100

- Large global workforce and active campus recruitment indicate strong need for scalable assessment solutions.
- Existing integration with HackerEarth shows openness to third-party platforms.
- Internal assessment tools and unclear decision-maker landscape reduce immediate urgency.

Key takeaways

- Their existing recruitment stack (Zoho Recruit) integrates with HackerEarth assessments, showing openness to third-party testing tools and a need for scalable campus-hiring solutions. [1][2]
- Key risk is that Zoho already has in-house assessment features and no publicly identified senior owners of campus hiring, which may slow adoption and decision-making. [3][4]

Snapshot

- The company operates in more than 80 countries with offices and data centers across Asia, Europe, and the United States. [5]
- Zoho runs its own applicant tracking system, Zoho Recruit, which supports candidate sourcing, workflow automation, and pre-screening assessments. [1][3]
- Zoho explicitly defines and engages in campus recruitment as a hiring channel. [6]

Live UI: plan, trace, metrics and a cited brief

Why you ask changes what you need

Same company — Zoho — three people, three different research jobs



Sales rep

"We sell a campus hiring platform. How do we pitch Zoho?"

Needs: Size, hiring programs, current tools, buyer roles, buying signals



Competitor analyst

"How does Freshworks compete with us?"

Needs: Products, pricing, positioning, recent moves, threat level



Interview candidate

"Prep me for an SDE intern interview at Zoho."

Needs: Tech stack, engineering culture, interview formats, recent news

Most research agents run one generic prompt and return one generic summary. **Scout turns the purpose into the plan.**

What Scout does

Four promises, each enforced in code — not just asked for in a prompt



Plans around the purpose

Intake classifies the goal (sales, competitor, interview prep); a playbook + past lessons shape a ≤ 5 -step plan.



Cites every claim

Every brief claim must point to evidence the agent actually collected with its tools.



Checks its own work

A critic grades each step (retry / follow-up / unknown); a deterministic verifier checks citations and numbers.



Learns with use

Verified facts, source reliability scores and strategy lessons carry into the next run.

Compare plans

Plan A

20260927-155410-bc87 · Sales prospect · Zoho

Plan B

20260927-163916-c8fa · Interview prep · Zoho

Sales prospect · Zoho

We sell a campus hiring-challenge platform. Research Zoho as a prospect and tell us how to pitch.

1. What is Zoho's overall company size, headquarters location, and global office footprint?
2. How many university recruitment programs, campus hiring events, or graduate-entry roles does Zoho run annually, and which campuses are targeted?
3. What talent acquisition tools, assessment platforms, or hiring-challenge solutions does Zoho currently use or mention in its tech stack?
4. Who are the senior decision-makers responsible for campus recruiting, talent acquisition, or early-career hiring at Zoho (roles only, not names)?
5. What recent news, product launches, funding rounds, or strategic initiatives has Zoho announced that could create a buying signal for a campus hiring-challenge platform?

Interview prep · Zoho

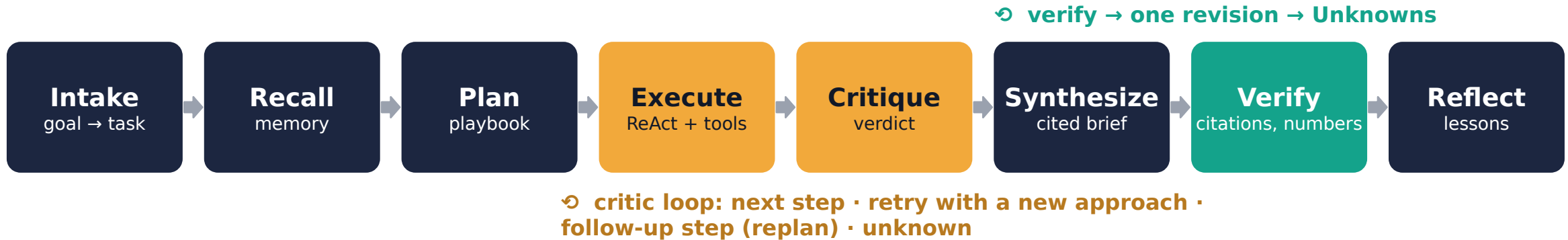
Prep me for an SDE intern interview at Zoho.

1. What are Zoho's documented recruitment process details, pre-screening assessment tools, and campus hiring approach relevant to SDE intern candidates? 🔥 from memory
2. What is Zoho's current technology stack for its core SaaS products (programming languages, frameworks, databases, cloud infrastructure, and CI/CD tools) as described in official engineering blogs or reputable tech articles?
3. What major news about Zoho has been published in the last 12 months (product launches, acquisitions, funding, partnerships, or engineering initiatives) from press releases or reputable tech news sites?
4. What does Zoho's engineering culture emphasize (e.g., development methodology, open-source contributions, internal tooling, remote work policy, learning programs) according to its engineering blog, developer portal, or employee testimonials?
5. What topics and question formats are commonly reported by recent candidates for Zoho SDE intern interviews (coding platforms used, problem types, system design depth, language focus)?

Same company, two purposes → two different plans (recorded runs 1 and 2). Run 2's first step was answered from memory.

Architecture: an explicit state machine

Hand-written, no agent framework. The orchestrator is a Python generator of events.



LLM + router

JSON calls validated by Pydantic, one repair; tiered router over 3 Groq models + Gemini, failover on quota/401/404



Tools

web_search (Tavily → DuckDuckGo), fetch_page (SSRF-guarded, trafilatura), recall_memory



Memory

SQLite: runs, facts (7-day / 2-day freshness), source scores, lessons, feedback



Interfaces

Rich CLI, Streamlit UI and trace.jsonl all consume the same event stream → free replay

Agentic patterns, mapped to code

Every pattern lives in a small, tested module



Plan-and-Execute

planner.py writes the plan;
orchestrator.py runs it



ReAct

executor.py: think →
search / fetch / recall →
observe



Tool use

tools/registry.py: name,
description, JSON schema
per tool



Critic routing

critic.py: complete · retry ·
follow-up · unknown



Self-verification

verifier.py: deterministic
checks + one revision



Reflexion lessons

reflector.py writes and
votes; top 3 go into the
next plan



Human feedback

👍/👎 per section updates
source scores and lesson
votes



Provider failover

router.py: ordered
candidates, cooldowns,
traced switches

It checks its own work

A critic after every step, a deterministic verifier before anything is published

```
▼ Step 2: retried → unknown · 6 tool calls · critic: retry (new approach: Search for "Zoho engineering blog database MySQL...")

What is Zoho's current technology stack for its core SaaS products (programming languages, frameworks, databases, cloud infrastructure, and CI/CD tools) as described in official engineering blogs or reputable tech articles?

🗨 I need to find Zoho's technology stack from official engineering blogs or reputable tech articles. Let me search for this information.

🔍 web_search(query='Zoho technology stack programming languages frameworks databases engineering blog', max_results=8)

👁 5 results

🗨 The search results are mostly generic tech stack articles, not Zoho-specific engineering content. Let me search more specifically for Zoho's engineering blog or tech articles about their actual stack.

🔍 web_search(query='Zoho engineering blog "Java" OR "Python" OR "Go" OR "Kotlin" technology stack data centers CI/CD', max_results=8)

👁 5 results

🗨 The search results are mostly generic tech-stack articles, not Zoho-specific engineering content. The ZDNET article and the Smartmates article look most promising for Zoho-specific stack details. Let me fetch the ZDNET article which discusses Zoho's full stack and infrastructure.

🔍 fetch_page(url='https://www.zdnet.com/article/zoho-full-stack-operating-system-and-data-protector/', focus='Zoho technology stack, programming languages, frameworks, databases, cloud infrastructure, data centers, CI/CD')

👁 1973 chars: To Zoho, this is how to deliver a seamless and end-to-end user experience (UX). Zoho is as much a cloud software company as it is a technolo

🗨 The ZDNET article confirms Zoho owns its data centers and builds all software in-house, but doesn't list specific languages/frameworks. The Tech Magify page mentions React, Python, Flutter, Node.js. I have enough from the search snippets and the fetched page to compile findings, though specific CI/CD tools and database names weren't explicitly confirmed in these sources. I'll report what's verifie...

🔗 Critic: retry — The executor failed to identify the primary backend language (Java) and specific CI/CD tools, relying on a low-quality aggregator (TechMagify) and a general news article instead of Zoho's official engineering blog or reputable technical deep-dives. | sources: https://catalyst.zoho.com/why-catalyst.html useless, https://www.techmagify.com/technology-landscape.php useless, https://www...

🔄 Retry with new approach: Search specifically for 'Zoho engineering blog Java' or 'Zoho backend technology stack Java Spring' and fetch pages from zoho.com/insights or zoho.com/blog that discuss their engineering practices, as Zoho is famously a Java-centric shop.

🗨 Search for Zoho's engineering blog or tech articles discussing their backend technology stack, specifically Java and related frameworks.
```

Critic (LLM, per step)

Rates each step and every source it saw. On weak evidence it orders a retry with a concrete new approach — e.g. “search the Zoho engineering blog for Java” instead of an aggregator.

Verifier (code, per brief)

Every claim must cite existing evidence, and every number in a claim must appear in a cited snippet. One LLM revision; what still fails moves to Unknowns.

100%

citation coverage

24

claims flagged by the verifier

7

moved to Unknowns, not published

Across 13 recorded runs (evals/results.md)

It gets faster with use

Same question asked twice — “Prep me for an SDE intern interview at Zoho.” — run 2 vs run 4

–33%

tool calls

12 → 8

–29%

LLM calls

31 → 22

–24%

tokens

40,040 → 30,478

–51%

time

203.7 s → 99.1 s

2×

facts reused

6 → 12



Fact memory

Verified facts per company, with source URL and freshness (7 days stable, 2 days news). Steps memory answers skip research but keep citations.



Source reliability

Critic + 👍/👎 ratings per domain: (useful+1)/(useful+useless+2). Search results are gently re-ranked.



Strategy lessons

The reflector writes 1–3 lessons per run; the top 3 per purpose shape the next plan and get voted on.

What carries over between runs

Four live runs recorded after a memory reset — committed in examples/



Run 1

Zoho · sales

Cold start. Researches size, campus programs, hiring tools, buyer roles. Writes lessons for sales prospects.

Run 2

Zoho · interview

Recalls 8 facts from run 1: memory answers step 1; research goes to new questions (stack, culture, formats).

Run 3

Freshworks · sales

No stored facts, but 2 lessons from run 1 shape the plan (news + investor-relations sources).

Run 4

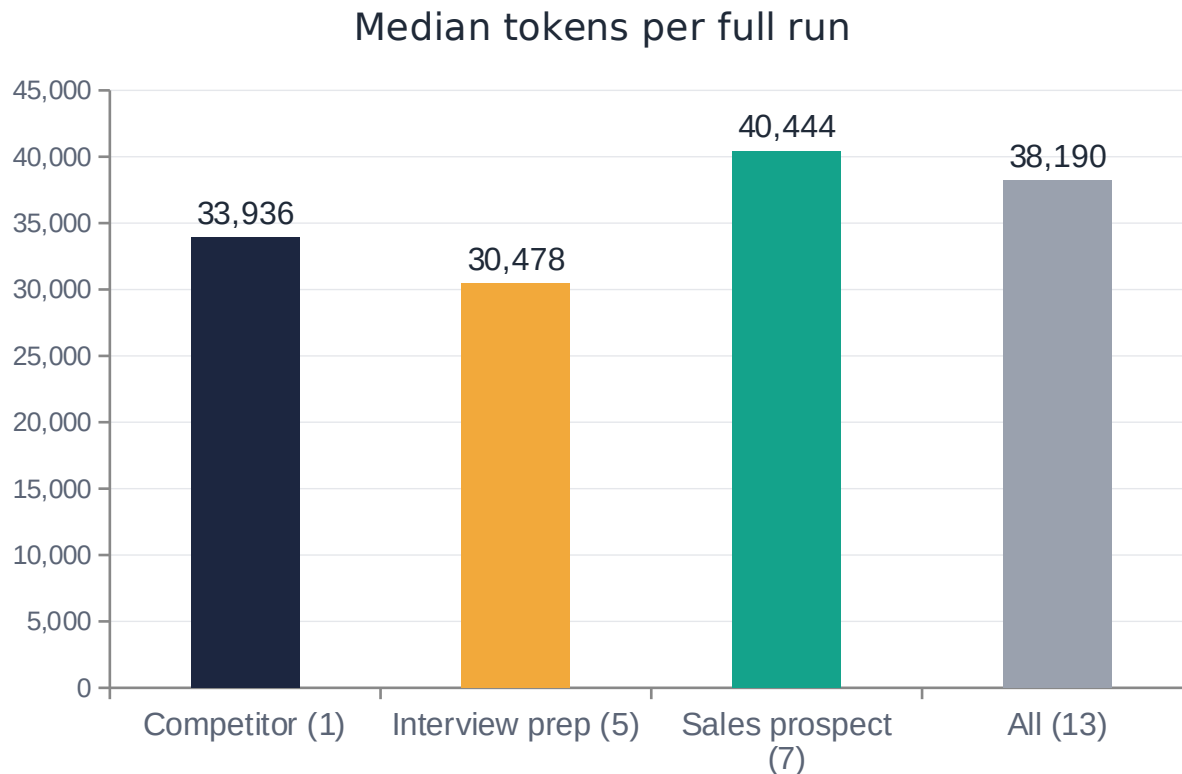
Zoho · interview again

Reuses 12 facts and run 2's 3 lessons: 8 tool calls instead of 12, half the time.

Lesson injected into run 3: *“Combine Google News with press-release aggregators and the company’s investor-relations page to capture funding rounds or strategic initiatives that signal buying intent.”*

Evals from real runs

python -m scout eval --from-runs · 13 recorded development runs, not a curated benchmark



Purpose	Runs	Median sec	Coverage	Budget stops
Competitor	1	148.2	100%	0
Interview prep	5	99.1	100%	2
Sales prospect	7	155.3	100%	1
All	13	137.0	100%	3

- 19k–58k tokens per full run on free tiers
- Budget stops end in a partial brief that says so — never a crash
- Live task set: evals/tasks.yaml (one task per purpose)

Reliability and safety

Built for the open web and free-tier APIs



Hard budgets

5 steps (8 with follow-up), 30 tool calls, 60 LLM calls, 480 s. Two LLM calls reserved so a brief always gets written.



SSRF guard

fetch_page allows http/https only, blocks private, loopback and link-local addresses, re-checks every redirect.



Untrusted content

All web text wrapped in `<untrusted_content>`; prompts treat it as data, never instructions.



Roles, not people

No personal profiles fetched or cited; emails and phone numbers scrubbed from briefs.



Provider failover

Cooldowns on quota errors, failover on 401/403/404, every switch traced; rate-limit waits shown live.



258 offline tests

Fake LLM and fake search, no network — including a Streamlit AppTest of the UI.

Design decisions and trade-offs

What we chose, why, and what it costs

Decision	Why	Cost
Hand-written state machine, no framework	Every edge, budget and failure is visible and unit-tested; replay falls out of the event stream	More code to own
JSON action protocol, not native tool calls	Portable across OpenAI-compatible providers; Pydantic-validated, one repair	Must also accept native-call shapes some models emit
SQLite memory	Zero setup after clone; transactional; inspectable	Single-user; no fuzzy (vector) recall
Deterministic checks behind the LLM	Citations, numbers, personal data and memory reuse don't depend on model quality	Exact-match numeric check can be strict
Free-tier multi-model router	Runs without paid keys	Rate limits slow runs; quality varies by model
Replay-first UI	Judging and demos cost no quota and are reproducible	Replays must be labelled (the UI does)

Honest limits, and what's next

Limitations

- Free-tier quotas cap live runs at a handful per day
- Planner quality varies by model
- Numeric check is exact-match after normalisation
- Lessons are not always fully company-neutral yet

Next

- Scheduled competitor tracking
- Vector memory for fuzzy fact recall
- Parallel execution of independent steps
- Scout as an MCP server



Try it in minutes — no API key needed to replay the example runs

github.com/UditSinghChauhan/scout-agent

```
git clone https://github.com/UditSinghChauhan/scout-agent.git && cd scout-agent
python3 -m venv .venv && source .venv/bin/activate && pip install -r requirements.txt
cp .env.example .env && python -m scout ui
```